

SQLite3.py

```
1 import sqlite3
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 from sqlite3database.helper import connect
5
6 connection=sqlite3.connect('sales_data.db')
7
8 cursor=connection.cursor()
9
10 # for creating the table
11 command="""create table if not exists
12 sales(p_id integer primary key,p_name text,quantity integer,price numeric)"""
13
14 cursor.execute(command)
15
16 #for inserting values
17 cursor.execute("insert into sales values(101,'tv',2,45908.00)")
18 cursor.execute("insert into sales values(102,'ac',3,67583.00)")
19 cursor.execute("insert into sales values(103,'tv',2,56987.00)")
20 cursor.execute("insert into sales values(104,'cooler',3,34897.00)")
21 cursor.execute("insert into sales values(105,'fan',4,45908.00)")
22 cursor.execute("insert into sales values(106,'mobile',4,78908.00)")
23 cursor.execute("insert into sales values(107,'ac',2,34987.00)")
24 cursor.execute("insert into sales values(108,'fan',2,8768.00)")
25 cursor.execute("insert into sales values(109,'cooler',1,4567.00)")
26 cursor.execute("insert into sales values(110,'mobile',2,56879.00)")
27 cursor.execute("insert into sales values(111,'tv',3,67893.00)")
28 cursor.execute("insert into sales values(112,'tv',2,45908.00)")
29 cursor.execute("insert into sales values(113,'ac',3,67583.00)")
30 cursor.execute("insert into sales values(114,'tv',2,56987.00)")
31 cursor.execute("insert into sales values(115,'cooler',3,34897.00)")
32 cursor.execute("insert into sales values(116,'fan',4,45908.00)")
```

1 1 ^ v

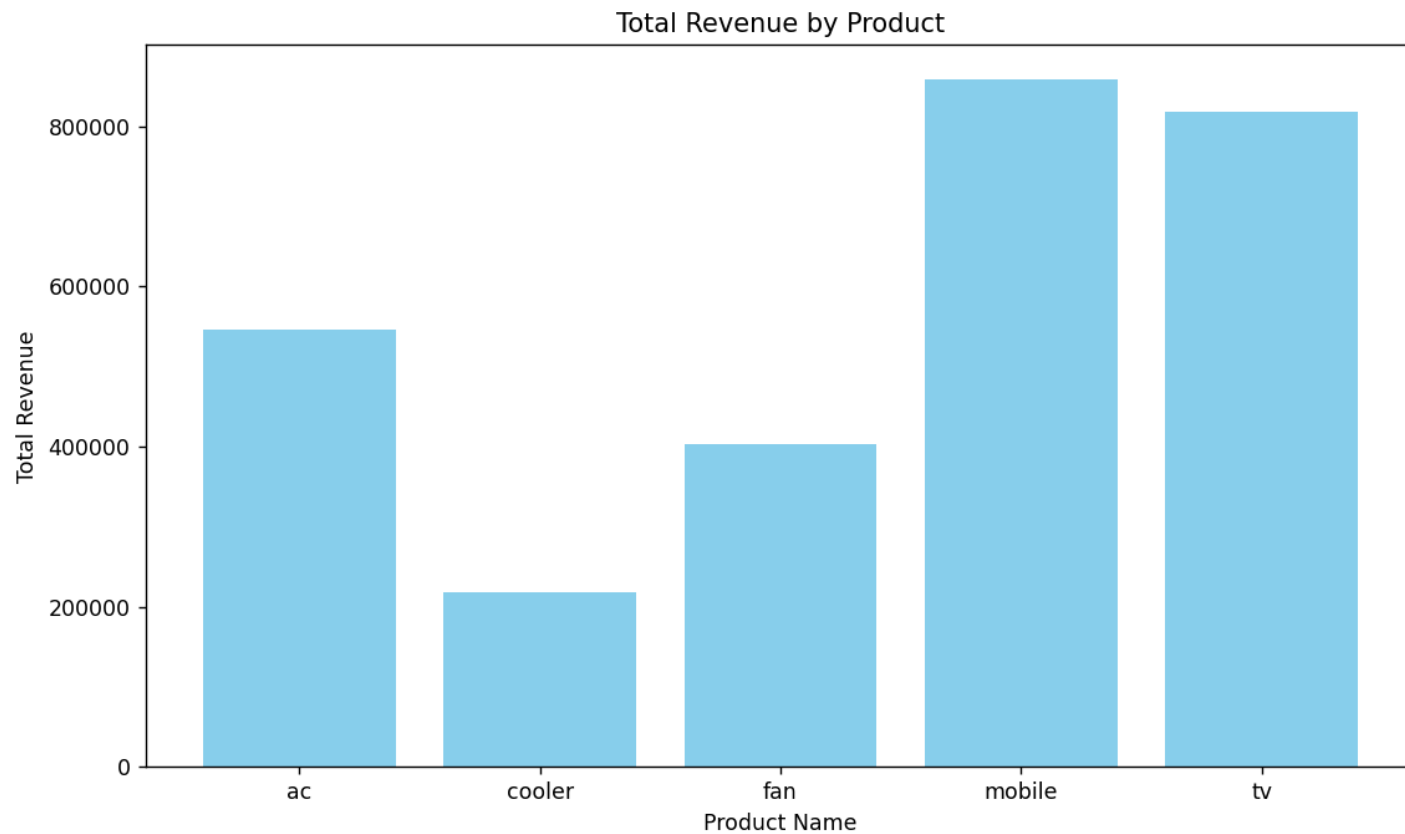
```
32 cursor.execute("insert into sales values(116,'fan',4,45908.00)")
33 cursor.execute("insert into sales values(117,'mobile',4,78908.00)")
34 cursor.execute("insert into sales values(118,'ac',2,34987.00)")
35 cursor.execute("insert into sales values(119,'fan',2,8768.00)")
36 cursor.execute("insert into sales values(120,'cooler',1,4567.00)")
37 cursor.execute("insert into sales values(121,'mobile',2,56879.00)")
38 cursor.execute("insert into sales values(122,'tv',3,67893.00)")
39 #get results
40 cursor.execute("select *from sales")
41
42 results1=cursor.fetchall()
43 print(results1)
44
45 df=pd.read_sql_query(sql: "select p_name as product_name,sum(quantity) as total_quantity, sum(quantity*price) as total_revenue from sales group by p_name",connection)
46 print(df)
47
48 plt.figure(figsize=(10, 6))
49 plt.bar(df['product_name'], df['total_revenue'], color='skyblue')
50 plt.xlabel('Product Name')
51 plt.ylabel('Total Revenue')
52 plt.title('Total Revenue by Product')
53 plt.show()
54
55 plt.figure(figsize=(8, 8))
56 plt.pie(df['total_quantity'], labels=df['product_name'], autopct='%1.1f%%', startangle=140)
57 plt.title('Total Quantity Distribution by Product')
58 plt.axis('equal') # Ensures pie is drawn as a circle
59 plt.tight_layout()
60 plt.show()
```



```
SQLite3.py x
34 cursor.execute("insert into sales values (110, 'ac', 3, 675834)")
35 cursor.execute("insert into sales values (111, 'ac', 3, 675834)")
36 cursor.execute("insert into sales values (112, 'ac', 3, 675834)")
37 cursor.execute("insert into sales values (113, 'ac', 3, 675834)")
38 cursor.execute("insert into sales values (114, 'ac', 3, 675834)")
39 #get results
40 cursor.execute("select *from sales")
41
42 results1=cursor.fetchall()
43 print(results1)
44
45 df=pd.read_sql_query(sql= "select p_name, total_quantity, total_revenue from sales")
46 print(df)
47
48 plt.figure(figsize=(10, 6))
49 plt.bar(df['product_name'], df['total_revenue'])
50 plt.xlabel('Product Name')
51 plt.ylabel('Total Revenue')
```

```
SQLite3 x
"C:\Users\ramak\Pandas Programs\.venv\Scripts"
[(101, 'tv', 2, 45908), (102, 'ac', 3, 67583), (103, 'fan', 12, 40233), (104, 'mobile', 12, 85878), (105, 'cooler', 8, 21851)]
product_name  total_quantity  total_revenue
0            ac              10         545440
1        cooler               8         218510
2            fan             12         402330
3        mobile             12         858780
4            tv             14         818930
```

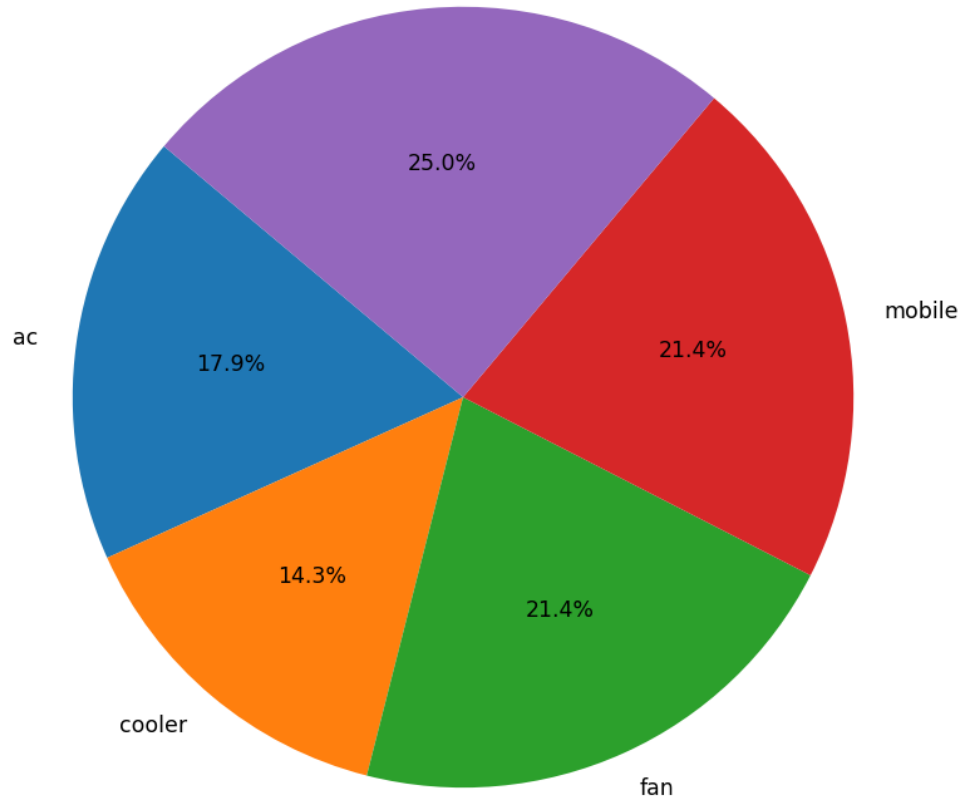
Figure 1



```
34 cursor.execute("insert into sales values(118,'ac',2,34987.00)")
35 cursor.execute("insert into sales values(119,'fan',2,8768.00)")
```

Figure 1

Total Quantity Distribution by Product



```
from sales group by p_name",connection)
```

```
mobile', 4, 78908), (107, 'ac', 2, 34987), (108,
```

```
"C:\Us
[(101,
prod
```

```
0
1
2
3
4
```